Experiment No- 4

Normalization

Normalization : Normalization in a database is the process of organizing data into tables to reduce redundancy and improve data integrity by dividing data into smaller, related tables and establishing relationships between them.

Forms of Normalization :

The forms of normalization in databases are structured steps to organize data effectively. Each form progressively refines the database by addressing specific problems like redundancy, dependency, and anomalies.

Here are the forms of normalization:

1. First Normal Form (1NF)
2. Second Normal Form (2NF)
3. Third Normal Form (3NF)
4. Boyce-Codd Normal Form (BCNF)
5. Fourth Normal Form (4NF)

1. First Normal Form (1NF) :

1. A relation will be 1NF if it contains an atomic value.

2. It states that an attribute of a table cannot hold multiple values. It must hold only single-valued attribute.

3. First normal form disallows the multi-valued attribute, composite attribute, and their combinations.

Example : - Relation EMPLOYEE is not in 1NF because of multi-valued attribute EMP_PHONE.

EMP_ID	EMP_NAME	EMP_PHONE	EMP_STATE
14	John	7272826385, 9064738238	UP
20	Harry	8574783832	Bihar
12	Sam	7390372389, 8589830302	Punjab

The decomposition of the EMPLOYEE table into 1NF has been shown below:

EMP_ID	EMP_NAME	EMP_PHONE	EMP_STATE
14	John	7272826385	UP
14	John	9064738238	UP
20	Harry	8574783832	Bihar
12	Sam	7390372389	Punjab
12	Sam	8589830302	Punjab

2. Second Normal Form (2NF):

1. In the 2NF, relational must be in 1NF.

2. In the second normal form, all non-key attributes are fully functional dependent on the primary key

Example –

Let's assume, a school can store the data of teachers and the subjects they teach. In a school, a teacher can teach more than one subject.

TEACHER_ID	SUBJECT	TEACHER_AGE
25	Biology	30
25	Chemistry	30
47	English	35
83	Computer	38
83	Math	38

In the given table, non-prime attribute TEACHER_AGE is dependent on TEACHER_ID which is a proper subset of a candidate key. That's why it violates the rule for 2NF.

To convert the given table into 2NF, we decompose it into two tables:

TEACHER_ID	TEACHER_AGE
25	30
47	35
83	38

TEACHER_ID	SUBJECT
25	Biology
25	Chemistry
47	English
83	Computer
83	Math

3.Third Normal Form (3NF)

1. A relation will be in 3NF if it is in 2NF and not contain any transitive partial dependency.
2. 3NF is used to reduce the data duplication. It is also used to achieve the data integrity.
3. If there is no transitive dependency for non-prime attributes, then the relation must be in third normal form.

Example –

EMP_ID	EMP_NAME	EMP_ZIP	EMP_STATE	EMP_CITY
222	Harry	201010	UP	Noida
333	Stephan	02228	US	Boston
444	Lan	60007	US	Chicago
555	Katharine	06389	UK	Norwich
666	John	462007	MP	Bhopal

Super key in the table above:

{EMP_ID}, {EMP_ID, EMP_NAME}, {EMP_ID, EMP_NAME, EMP_ZIP}....so on

Candidate key: {EMP_ID}

Non-prime attributes: In the given table, all attributes except EMP_ID are non-prime.

Here, EMP_STATE & EMP_CITY dependent on EMP_ZIP and EMP_ZIP dependent on EMP_ID. The non-prime attributes (EMP_STATE, EMP_CITY) transitively dependent on super key(EMP_ID). It violates the rule of third normal form.The reduction of 2NF relation into 3NF consists of splitting the 2NF into appropriate relations such that every non-key attribute are functionally dependent on the primary key not transitively or indirectly of the respective relations.

That's why we need to move the EMP_CITY and EMP_STATE to the new <EMPLOYEE_ZIP> table, with EMP_ZIP as a Primary key.

EMP_ID	EMP_NAME	EMP_ZIP
222	Harry	201010
333	Stephan	02228
444	Lan	60007
555	Katharine	06389
666	John	462007

EMP_ZIP	EMP_STATE	EMP_CITY
02228	US	Boston
06389	UK	Norwich
201010	UP	Noida
462007	MP	Bhopal
60007	US	Chicago

4. Boyce-Codd Normal Form (BCNF):

1. BCNF is the advance version of 3NF. It is stricter than 3NF.

2. A table is in BCNF if every functional dependency X → Y, X is the super key of the table.

3. For BCNF, the table should be in 3NF, and for every FD, LHS is super key.

Example -

Let's assume there is a company where employees work in more than one department.

EMP_ID	EMP_COUNTRY	EMP_DEPT	DEPT_TYPE	EMP_DEPT_NO
264	India	Designing	D394	283
264	India	Testing	D394	300
364	UK	Developing	D283	549
364	UK	Stores	D283	232

In the above table Functional dependencies are as follows:

1. EMP_ID → EMP_COUNTRY
2. EMP_DEPT → {DEPT_TYPE, EMP_DEPT_NO}

Candidate key: {EMP-ID, EMP-DEPT}

The table is not in BCNF because neither EMP_DEPT nor EMP_ID alone are keys.

To convert the given table into BCNF, we decompose it into three tables:

EMP_ID	EMP_COUNTRY
264	India
264	India

EMP_DEPT	DEPT_TYPE	EMP_DEPT_NO
Designing	D394	283
Testing	D394	300
Stores	D283	232
Developing	D283	549

EMP_DEPT	DEPT_TYPE	EMP_DEPT_NO
Designing	D394	283
Testing	D394	300
Stores	D283	232
Developing	D283	549